

ROHIT RAVINDRA V.

Pune, Maharashtra

+91 7579415078 | rohitvalsetwar@gmail.com | [LinkedIn](#) | [LeetCode](#) | [GitHub](#) | [My Portfolio](#)

EDUCATION

Vishwakarma Institute of Technology — CGPA: 7.8/10

Bachelor of Technology, Mechanical Engineering

Aug 2023 – Jun 2027

Maharashtra, India

Army Public School, Ambala Cantt — Class XII (PCM + AI)

Haryana, India

Apr 2022 – Jan 2023

Army Public School, Kamptee — Class X

Maharashtra, India

Mar 2021

MAJOR PROJECTS

Quick Bites – AI-Powered Food Delivery Platform

Mar 2026 – Jul 2026

MERN Stack (MongoDB, Express.js, React.js, Node.js), Socket.io, Web Speech API

- Engineered a full-stack food delivery platform (Zomato-inspired) with 4 distinct AI-driven personalization modules.
- Formulated an AI Meal Recommender combining a temperature slider, a budget slider, and past order history to auto-suggest dishes — e.g., recommending cold coffee and mocktails above 35°C
- Curated a Mood-Based Food Selection engine mapping 4 moods (Happy, Study, Gym, Late Night) to targeted menu categories.
- Architected a Smart Group Ordering system using 6-digit join codes and a live shared cart, consolidating individual orders from up to 10 members into a single host-checkout.
- Deployed Voice Search via browser-based speech recognition on the restaurant listing page.

Steer-by-Wire System for 4-Wheeler Vehicles

Sept 2025 – Dec 2025

Embedded Systems, Sensor Integration, Mechanical Design, Arduino/ESP

- Designed and prototyped a Steer-by-Wire system replacing conventional mechanical steering linkage with electronically controlled actuation, cutting mechanical linkage components by 10%.
- Integrated angle and torque sensors with a motor-driven actuator to deliver real-time steering response within 300 ms.
- Embedded with 2 modes for parking and highway mode having ratio 1:1 and 1:0.5.

Marine Communication System (Defence Use)

Sept 2023 – Mar 2024

Maharashtra, India

- Developed a wireless, satellite-independent communication model enabling secure data exchange over a range of 500 meters for defence applications as a prototype.
- IEEE published Research paper

Heart Attack Prediction System

Jan 2025 – Jul 2025

Maharashtra, India

- Trained an AI/statistical model to forecast cardiac events up to 5 minutes in advance, achieving 42% prediction accuracy on a dataset of patient records

Non-Destructive Rust Detection System

Aug 2025 – Dec 2025

Maharashtra, India

- Prototyped a non-destructive testing system using magnetic flux leakage as an input feature to an AI model, detecting corrosion in metals with 94% accuracy
- Earned an A+ evaluation from faculty and positive feedback from industry professionals; patent published.

Headlight Dimming System for 4-Wheelers

Feb 2025 – Jun 2025

India

- Built an IoT-based headlight dimming prototype that automatically reduces high-beam glare, validated across 3 simulated driving scenarios

SEMI-MAJOR PROJECTS

LuxNest – Hotel Booking Platform

Node.js, Express.js, MongoDB, EJS, Bootstrap

- Developed a full-stack hotel booking web application with secure user authentication and session management
- Implemented CRUD operations across listings, bookings, and user profiles, supporting a catalog of 70+ listings
- Styled a responsive UI with Bootstrap, improving cross-device usability.
- Integrated REST APIs to enable seamless backend-frontend communication

Music Genre Prediction (ML/DL Project)

- Constructed a CNN-based deep learning model to classify music across 8 genre categories, achieving 90% test accuracy
- Processed audio data using feature extraction techniques (MFCC, spectrograms) across a dataset of 70,000 audio samples

Smart Loan Approval System

- Formulated a machine learning model to predict loan approval using EDA and feature engineering on a dataset.
- Applied classification algorithms (Logistic Regression, Decision Trees), achieving 94% model accuracy
- Enhanced accuracy by 2 percentage points through data preprocessing and hyperparameter tuning

CERTIFICATIONS

- Data Structures & Algorithms with Java
- MERN Web Development
- AI and Generative AI Certification (Ongoing)

TECHNICAL SKILLS

- Languages: Python, Java (DSA), JavaScript
- Web Development: React.js, Node.js, Express.js, Redux Toolkit, HTML, CSS, Bootstrap
- Databases: MongoDB, SQL
- Core Concepts: Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP), Operating Systems (in progress)
- AI/ML: Machine Learning, Deep Learning (Basics), NLP, Data Analysis, Data Science; GenAI, RAG, Transformers (in progress)
- Libraries & Frameworks: Scikit-learn, OpenCV, Pandas, NumPy, Matplotlib, Seaborn
- Techniques: Regression, Classification, Feature Engineering, Audio Processing, Text Processing
- IoT & Embedded: ESP, Embedded C, Arduino IDE, Sensor Integration
- DevOps (in progress): CI/CD Pipelines, Docker, Kubernetes